# TQP3M9037

## Ultra Low-Noise, High Linearity LNA

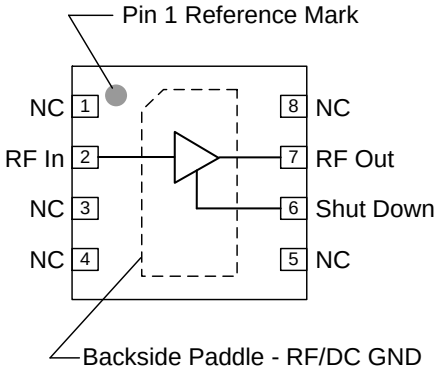
## General Description

The TQP3M9037 is a high-linearity, ultra-low noise gain block amplifier in a small 2 x 2 mm surface-mount package. At 1.95 GHz, the amplifier typically provides 20 dB gain, +35 dBm OIP3, and 0.4 dB noise figure while drawing 70 mA current from a +5 V supply. This amplifier does not require a negative supply for operation and can be biased from a single positive supply ranging from +3.3 to +5 volts. The device is housed in a green/RoHS-compliant industry-standard 2 x 2 mm package.

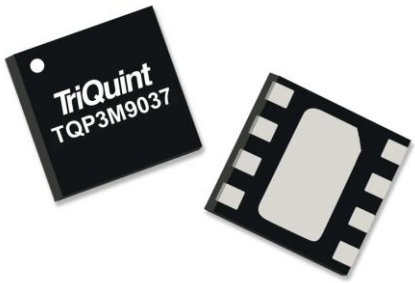
The TQP3M9037 is internally matched using a high performance E-pHEMT process and only requires four external components for operation from a single positive supply: an external RF choke and blocking/bypass capacitors. This low noise amplifier contains an internal active bias to maintain high performance over temperature and integrates a shut-down biasing capability to allow for operation for TDD applications.

The TQP3M9037 covers the 0.7 – 6.0 GHz frequency band and is targeted for wireless infrastructure. It is pin compatible with the low-band, 0.4–2.0 GHz TQP3M9036.

## Functional Block Diagram




Top View




8 Pin 2X2 mm DFN Package

## Product Features

- 0.7-6.0 GHz Operational Bandwidth
- Ultra Low Noise Figure, 0.4 dB NF at 1.95 GHz
- High Gain, 20 dB Gain at 1.95 GHz
- High Linearity, +35 dBm Output IP3
- High Input Power Ruggedness, +22 dBm CW
- Unconditionally Stable
- Integrated On-chip Matching, 50 Ohm In/Out
- Integrated Active Bias
- Integrated Shutdown Control Pin
- +3V to +5V Supply; Does Not Require −Vgg
- Pin Compatible with Low-Band TQP3M9036

## Applications

- Repeaters
- Mobile Infrastructure
- LTE / WCDMA / CDMA / GSM
- General Purpose Wireless
- TDD or FDD Systems

## Ordering Information

| Part No. | Description |
|---|---|
| TQP3M9037 | Ultra low noise, High IP3 LNA |
| TQP3M9037-PCB | 0.7 – 6.0 GHz Evaluation Board |

Standard T/R size = 2500 pieces on a 7” reel

## Absolute Maximum Ratings

| Parameter | Rating |
|---|---|
| Storage Temperature | −65 to 150°C |
| Supply Voltage ($V_{DD}$) | +7 V |
| RF Input Power, CW, 50Ω, T=25°C | +22 dBm |

Operation of this device outside the parameter ranges given above may cause permanent damage.

## Recommended Operating Conditions

| Parameter | Min | Typ | Max | Units |
|---|---|---|---|---|
| Supply Voltage ($V_{DD}$) | 3.0 | 4.2 | 5.25 | V |
| $T_{CASE}$ | −40 | | +105 | °C |
| Tj for >$10^6$ hours MTTF | | | +190 | °C |

Electrical specifications are measured at specified test conditions. Specifications are not guaranteed over all recommended operating conditions.

## Electrical Specifications

Test conditions unless otherwise noted: $V_{DD}$ =+5V, Temp=+25°C, 50 Ω system.

| Parameter | Conditions | Min | Typ | Max | Units |
|---|---|---|---|---|---|
| Operational Frequency Range | | 700 | | 6000 | MHz |
| Test Frequency | | | 1950 | | MHz |
| Gain | | 18.5 | 20.0 | 21.5 | dB |
| Input Return Loss | | | 10 | | dB |
| Output Return Loss | | | 11 | | dB |
| Noise Figure | | | 0.40 | 0.65 | dB |
| Output P1dB | | | +20 | | dBm |
| Output IP3 | Pout=+5 dBm/tone, Δf=1 MHz | +32 | +35 | | dBm |
| Power Shutdown Control ($V_{PD}$)[1] | On state | 0 | | 0.4 | V |
| | Off state (Power down) | 2.5 | 3.3 | $V_{DD}$ | V |
| Current, $I_{DD}$ | On state | 40 | 70 | 90 | mA |
| | Off state (Power down) | | 3 | 4 | mA |
| Shutdown pin current, $I_{SD}$ | $V_{PD}$ ≥ +2.5 V | | 140 | | µA |
| Switching Speed[(2)] | Rise time (10% to 90%) | | 116 | | ns |
| | Fall time (90% to 10%) | | 340 | | ns |
| Thermal Resistance, $\theta_{jc}$ | channel to case | | | 53.4 | °C/W |

Notes:
1. This voltage is referred to the turret labelled PD on the Qorvo EVB as shown on pg. 4.
2. Input DC block capacitor, C1 = 10 pF. Also see bottom of pg. 5.

## VPD (pin 6) Voltage Limits

| $V_{PD}$ | Min | Max | Units |
|---|---|---|---|
| $V_{low}$ | 0 | 0.1 | V |
| $V_{high}$ | 0.55 | $V_{DD}$ | V |

## S-Parameters

Test Conditions: $V_{DD}$=+5 V, $I_{DD}$=70 mA (typ.), T=+25°C, unmatched 50 ohm system, calibrated to device leads

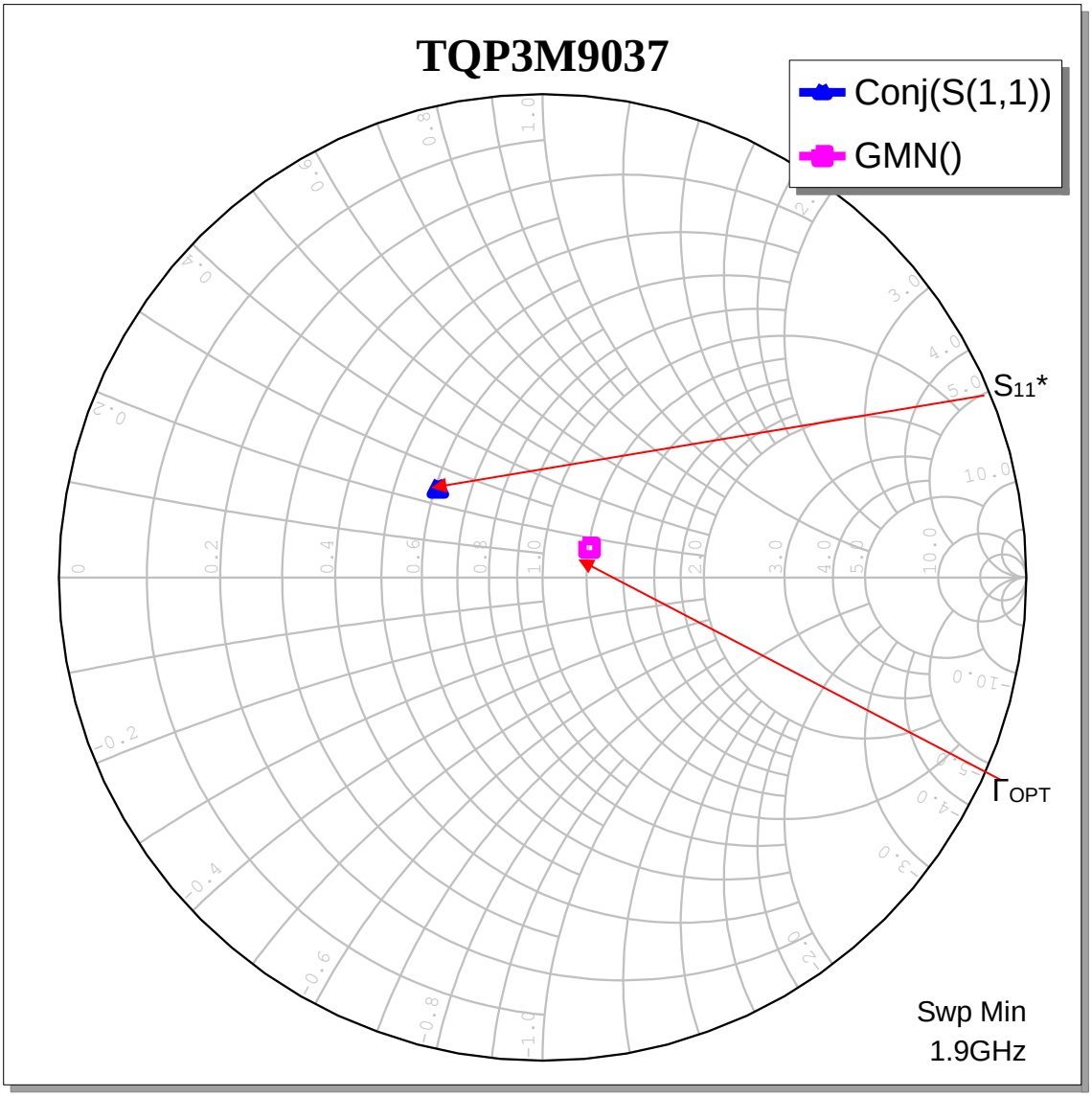
| Freq (GHz) | S11 (dB) | S11 (ang) | S21 (dB) | S21 (ang) | S12 (dB) | S12 (ang) | S22 (dB) | S22 (ang) |
|---|---|---|---|---|---|---|---|---|
| 0.05 | -7.08 | -13.38 | 28.74 | 173.92 | -32.91 | 8.58 | -15.03 | 0.20 |
| 0.10 | -7.30 | -16.67 | 28.70 | 169.79 | -32.76 | 4.97 | -14.12 | 2.20 |
| 0.20 | -7.55 | -27.25 | 28.51 | 160.47 | -32.60 | 5.69 | -13.67 | -1.25 |
| 0.40 | -7.89 | -48.81 | 27.68 | 143.44 | -32.47 | 8.81 | -13.09 | -6.33 |
| 0.60 | -8.28 | -67.86 | 26.67 | 128.94 | -32.31 | 11.76 | -12.61 | -12.97 |
| 0.80 | -8.67 | -82.83 | 25.57 | 116.72 | -31.94 | 14.83 | -12.17 | -19.29 |
| 1.00 | -9.10 | -96.01 | 24.54 | 106.22 | -31.49 | 16.95 | -11.87 | -25.85 |
| 1.20 | -9.48 | -106.85 | 23.55 | 97.05 | -30.96 | 17.84 | -11.62 | -31.76 |
| 1.40 | -9.97 | -116.61 | 22.63 | 88.79 | -30.34 | 19.60 | -11.45 | -37.19 |
| 1.60 | -10.36 | -126.05 | 21.78 | 81.19 | -29.81 | 19.78 | -11.41 | -42.80 |
| 1.80 | -10.78 | -135.06 | 20.99 | 74.13 | -29.42 | 19.53 | -11.43 | -47.57 |
| 2.00 | -11.14 | -144.52 | 20.28 | 67.42 | -28.85 | 19.14 | -11.56 | -53.18 |
| 2.20 | -11.43 | -153.91 | 19.61 | 60.92 | -28.45 | 19.04 | -11.79 | -58.28 |
| 2.40 | -11.54 | -163.77 | 19.00 | 54.49 | -27.99 | 17.71 | -12.17 | -64.71 |
| 2.60 | -11.50 | -174.25 | 18.41 | 48.28 | -27.67 | 16.22 | -12.63 | -71.10 |
| 2.80 | -11.30 | 175.64 | 17.86 | 41.94 | -27.19 | 14.85 | -13.20 | -79.45 |
| 3.00 | -10.90 | 166.39 | 17.33 | 35.76 | -26.85 | 12.36 | -13.86 | -89.04 |
| 3.20 | -10.45 | 157.46 | 16.79 | 29.69 | -26.50 | 10.04 | -14.51 | -101.79 |
| 3.40 | -9.89 | 150.34 | 16.29 | 23.57 | -26.19 | 8.06 | -14.74 | -116.68 |
| 3.60 | -9.29 | 144.41 | 15.76 | 17.56 | -25.95 | 5.99 | -14.57 | -133.42 |
| 3.80 | -8.86 | 140.00 | 15.22 | 11.78 | -25.75 | 3.39 | -13.78 | -148.27 |
| 4.00 | -8.32 | 136.87 | 14.67 | 6.13 | -25.57 | 1.64 | -12.70 | -161.52 |
| 4.20 | -8.03 | 133.99 | 14.10 | 0.74 | -25.50 | -1.15 | -11.25 | -169.83 |
| 4.40 | -7.68 | 136.01 | 13.50 | -4.52 | -25.43 | -2.83 | -10.05 | -178.99 |
| 4.60 | -7.40 | 136.69 | 12.83 | -9.19 | -25.38 | -4.24 | -8.64 | 175.79 |
| 4.80 | -7.21 | 140.78 | 12.13 | -13.41 | -25.53 | -5.31 | -7.25 | 170.49 |
| 5.00 | -6.35 | 149.93 | 11.26 | -16.91 | -25.78 | -6.38 | -6.41 | 163.10 |

## Noise Parameters

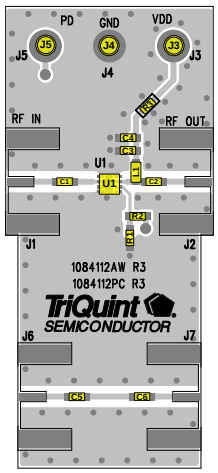
Test conditions unless otherwise noted: $V_{DD}$=+5 V, $I_{DD}$=70 mA (typ.), Temp=+25°C, 50 Ohm system

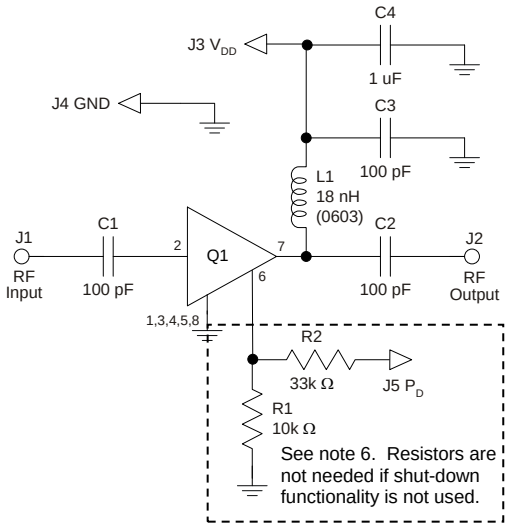
| Freq (GHz) | $NF_{min}$ (dB) | MagOpt (mag) | AngOpt (deg) | Rn (Ω) |
|---|---|---|---|---|
| 1.5 | 0.40 | 0.15 | 14.1 | 0.08 |
| 1.7 | 0.29 | 0.16 | 43.7 | 0.06 |
| 1.8 | 0.3 | 0.13 | 55.1 | 0.06 |
| 1.9 | 0.30 | 0.11 | 30.2 | 0.06 |
| 2 | 0.32 | 0.11 | 65.3 | 0.05 |
| 2.1 | 0.32 | 0.13 | 77.6 | 0.05 |
| 2.2 | 0.35 | 0.14 | 89.0 | 0.05 |
| 2.4 | 0.43 | 0.06 | 85.8 | 0.05 |
| 2.5 | 0.51 | 0.02 | 42.1 | 0.06 |
| 2.6 | 0.46 | 0.08 | 157.4 | 0.04 |
| 2.7 | 0.46 | 0.07 | 148.6 | 0.05 |

## Γopt and S11*



## TQP3M9037-PCB Evaluation Board




Notes:

1. See Evaluation Board PCB Information section for material and stack-up.
2. R3 (0 Ω jumper) is not shown on the schematic and may be replaced with copper trace in the target application layout.
3. All components are of 0402 size unless stated on the schematic.
4. C1, C2, and C3 are non-critical values. The reactive impedance should be as low as possible at the frequency of operation for optimal performance.
5. The L1 value is non-critical and needs to provide high reactive impedance at the frequency of operation.
6. R1 and R2 are optional and do not need to be loaded if the shut-down functionality is not needed; i.e. FDD applications. If R1 and R2 are not loaded, the LNA will operate in its standard "ON" state.
7. A through line is included on the evaluation board to de-embed the board losses.

## Bill of Material − TQP3M9037-PCB

| Reference Des. | Value | Description | Manuf. | Part Number |
|---|---|---|---|---|
| N/A | N/A | Printed Circuit Board | Qorvo | |
| U1 | n/a | Ultra Low Noise, High Linearity LNA | Qorvo | TQP3M9037 |
| R1 | 10 kΩ | Resistor, Chip, 0402, 5%, 1/16W | various | |
| R2 | 33 kΩ | Resistor, Chip, 0402, 5%, 1/16W | various | |
| R3 | 0 Ω | Resistor, Chip, 0402, 5%, 1/16W | various | |
| L1 | 18 nH | Inductor, 0603, 5%, Ceramic | various | |
| C4 | 1.0 uF | Cap., Chip, 0402, 10%, 10V, X5R | various | |
| C1, C2, C3, C5, C6 | 100 pF | Cap., Chip, 0402, 5%, 50V, NPO/COG | various | |

## Resistive Divider Values

| $V_{PD}$ Signal Level | +3.3 V TTL | +2.5 V TTL | Units |
|---|---|---|---|
| R1 | 10 kΩ | 11 kΩ | Ω |
| R2 | 33 kΩ | 33 kΩ | Ω |

## Switching Speed

$V_{PD}$ toggled with a 1KHz square wave(0-2V), $V_{DD}$=5V, RF freq = 1.95GHz

| Parameter | C1=100pF | C1=10pF | Units |
|---|---|---|---|
| Turn-off Fall time (50% TTL – 10% RF) | 1.4 | 0.34 | µs |
| Turn-on Rise time (50% TTL – 90% RF) | 0.39 | 0.116 | µs |

## Typical Performance – TQP3M9037-PCB $V_{DD}$ = +5 V

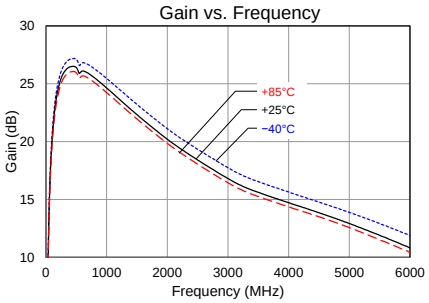
Test conditions unless otherwise noted: $V_{DD}$=+5 V, $I_{DD}$=70 mA (typ.), Temp=+25 °C

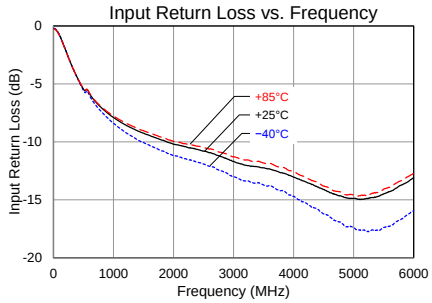
| Parameter | Conditions | Typical Values | | | | | Units |
|---|---|---|---|---|---|---|---|
| Frequency | | 900 | 1700 | 1950 | 2300 | 2600 | MHz |
| Gain | | 25.3 | 21.4 | 20.4 | 19.1 | 18.1 | dB |
| Input Return Loss | | 7.4 | 9.6 | 10.1 | 10.5 | 10.9 | dB |
| Output Return Loss | | 11.7 | 11.6 | 11.0 | 10.2 | 9.8 | dB |
| Output P1dB | | +22 | +20.7 | +20 | +19.8 | +19.5 | dBm |
| OIP3 | Pout=+5 dBm/tone, Δf=1 MHz | +34.4 | +34.9 | +35.7 | +35.1 | +35.0 | dBm |
| Noise figure (1) | | 0.4 | 0.36 | 0.41 | 0.43 | 0.53 | dB |

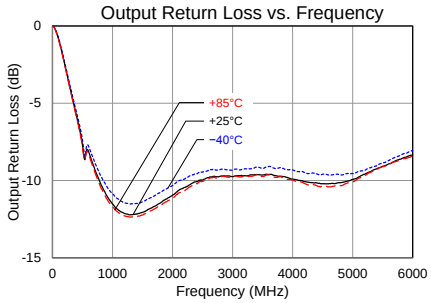
Notes:
1. Noise figure data shown in the table above is de-embedded to the device pin pads.

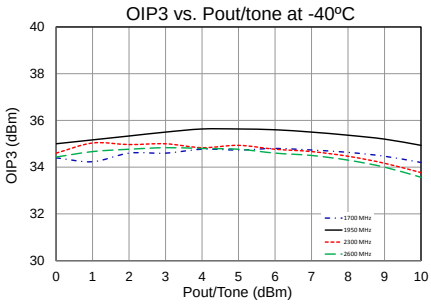
## Performance Plots – TQP3M9037-PCB $V_{DD}$ = +5 V

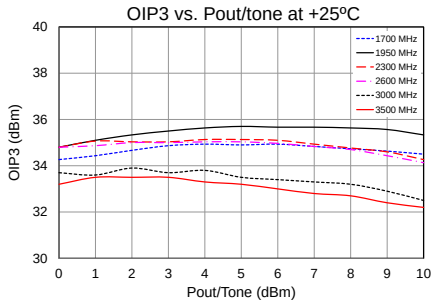
Test conditions unless otherwise noted: $V_{DD}$=+5 V, $I_{DD}$=70 mA (typ.), Temp=+25°C

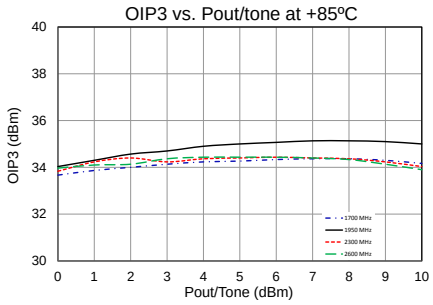


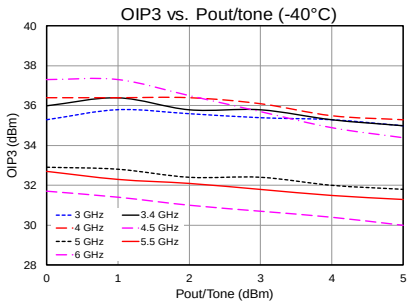



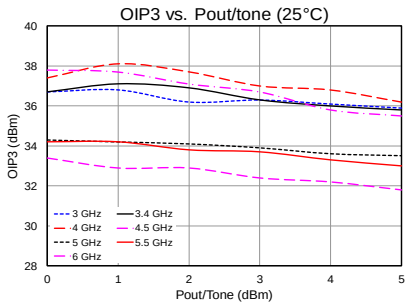



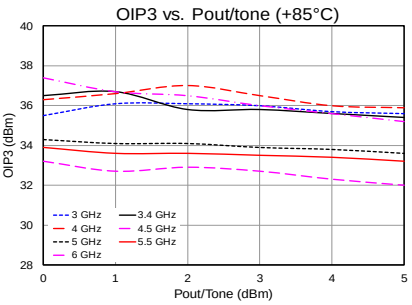



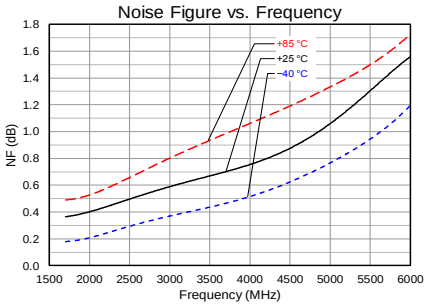



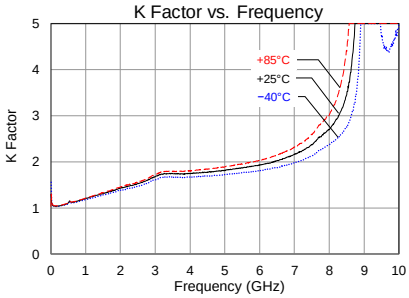



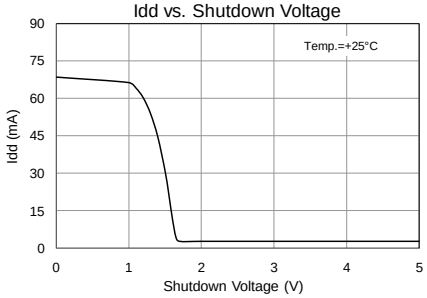



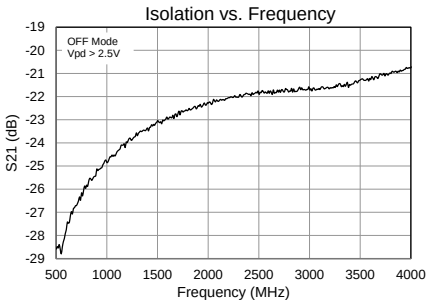
## Performance Plots − TQP3M9037-PCB $V_{DD}$ = +5 V

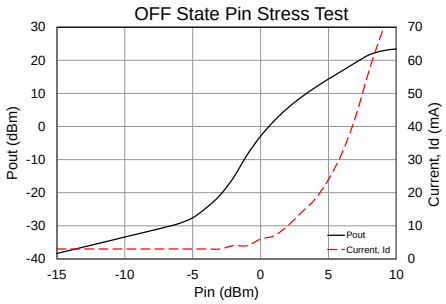
Test conditions unless otherwise noted: $V_{DD}$=+5 V, $I_{DD}$=70 mA (typ.), Temp=+25°C



## Typical Performance – TQP3M9037-PCB $V_{DD}$ = +3.3 V

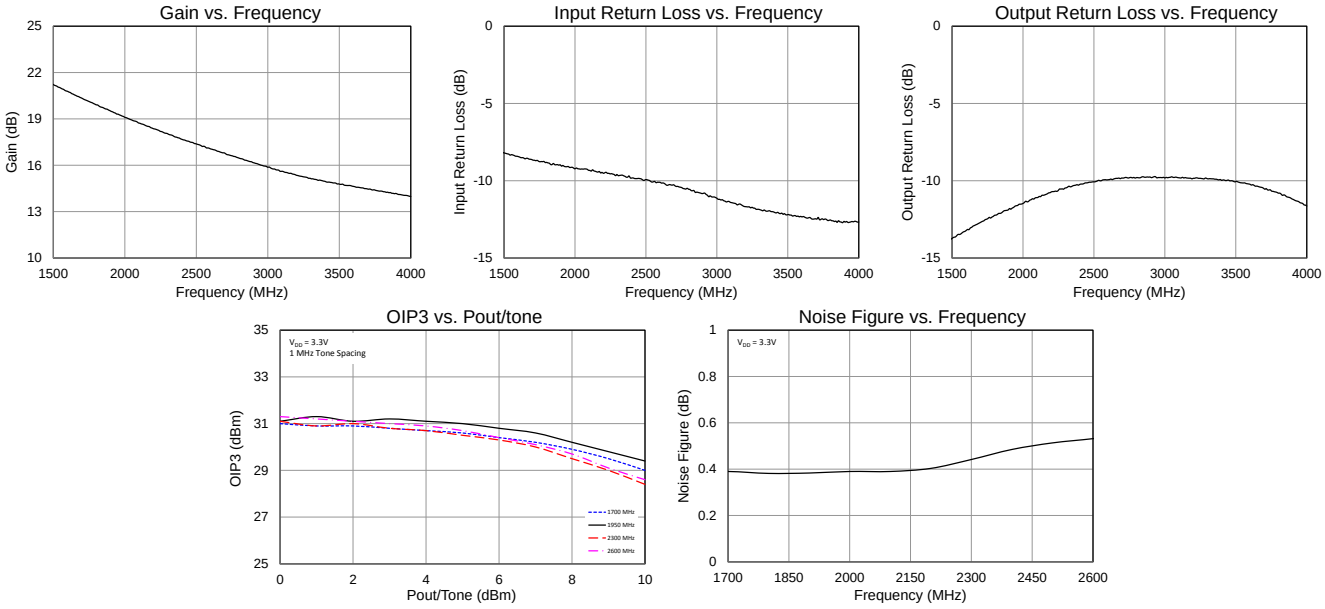
Test conditions unless otherwise noted: $V_{DD}$ = +3.3V, $I_{DD}$ = 44 mA (typ.), Temp= +25°C

| Parameter | Conditions | Typical Values | | | | Units |
|---|---|---|---|---|---|---|
| Frequency | | 1700 | 1950 | 2300 | 2600 | MHz |
| Gain | | 19.9 | 19.2 | 17.7 | 17.0 | dB |
| Input Return Loss | | 8.3 | 9.2 | 9.5 | 10.5 | dB |
| Output Return Loss | | 11.3 | 11.3 | 9.5 | 9.5 | dB |
| Output P1dB | | +16.2 | +16.1 | +15.5 | +15.5 | dBm |
| OIP3 | Pout = +5 dBm/tone, Δf=1 MHz | +30.6 | +31 | +30.5 | +30.7 | dBm |
| Noise figure (1) | | 0.4 | 0.38 | 0.44 | 0.53 | dB |

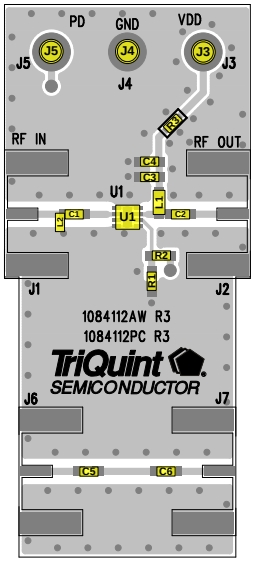
Notes:
1. Noise figure data shown in the table above is de-embedded from the eval board loss.

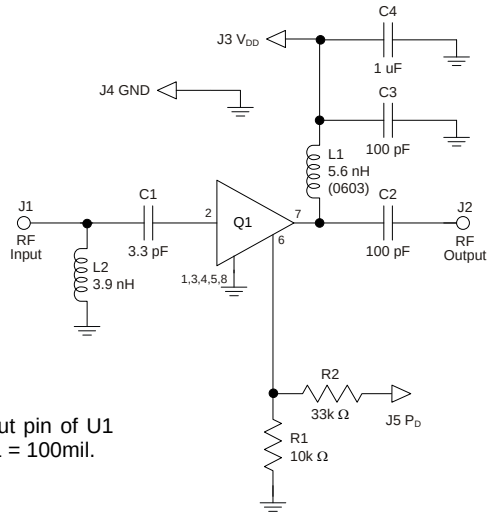
## Performance Plots – TQP3M9037-PCB $V_{DD}$ = +3.3 V

Test conditions unless otherwise noted: $V_{DD}$ = +3.3V, $I_{DD}$ = 44 mA (typ.), Temp= +25°C



## Reference Design – Optimized Return Loss




Notes:

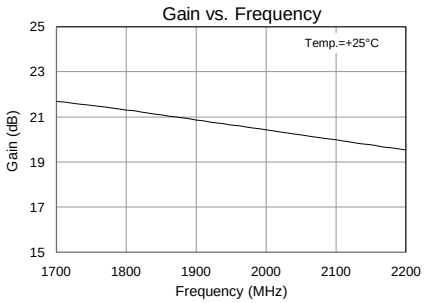
1. Distance from input pin of U1 to right edge of C1 = 100mil.

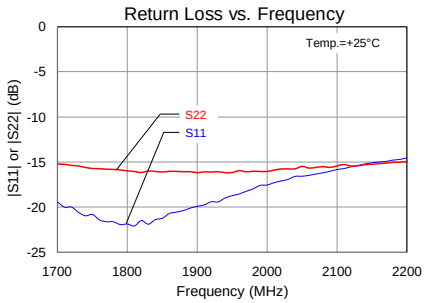
## Bill of Material

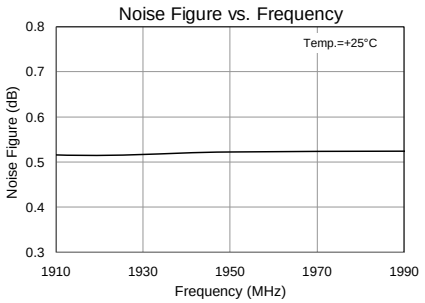
| Reference Des. | Value | Description | Manuf. | Part Number |
|---|---|---|---|---|
| n/a | n/a | Printed Circuit Board | Qorvo | |
| U1 | n/a | Ultra Low Noise Amplifier | Qorvo | TQP3M9037 |
| R1 | 10 kΩ | Resistor, Chip, 0402, 5%, 1/16W | various | |
| R2 | 33 kΩ | Resistor, Chip, 0402, 5%, 1/16W | various | |
| R3 | 0 Ω | Resistor, Chip, 0402, 5%, 1/16W | various | |
| L1 | 5.6 nH | Inductor, 0603, 5%, Ceramic | various | |
| L2 | 3.9 nH | Inductor, 0402, 5%, Ceramic | various | |
| C1 | 3.3 pF | Cap, Chip, 0402, ±0.25 pF, 50V, NPO/COG | various | |
| C4 | 1.0 uF | Cap, Chip, 0402, 10%, 10V, X5R | various | |
| C2, C3, C5, C6 | 100 pF | Cap, Chip, 0402, 5%, 50V, NPO/COG | various | |

## Performance Plots

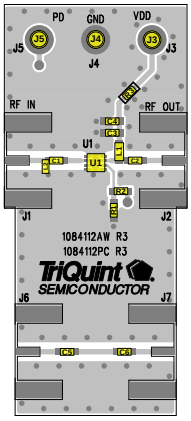
Test conditions unless otherwise noted: $V_{DD}$ = +5V, $I_{DD}$ = 70 mA (typ.), Temp= +25°C

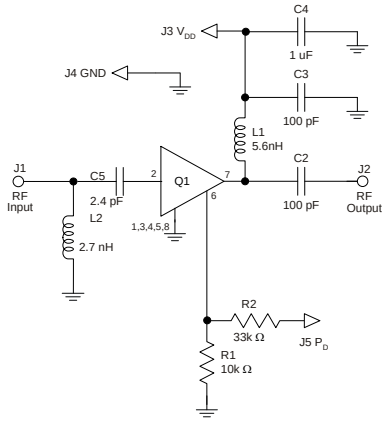


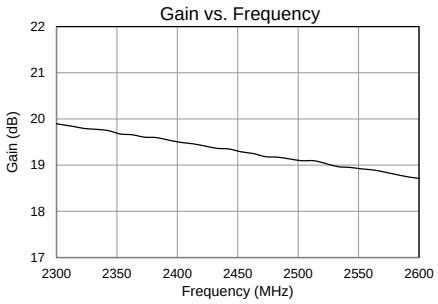



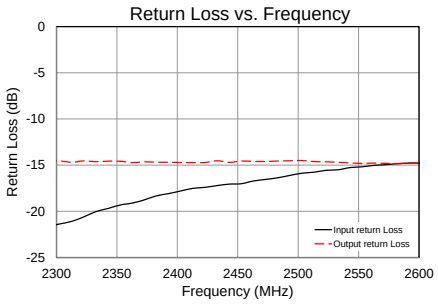
## Evaluation Board: 2300 – 2600 MHz Reference Design

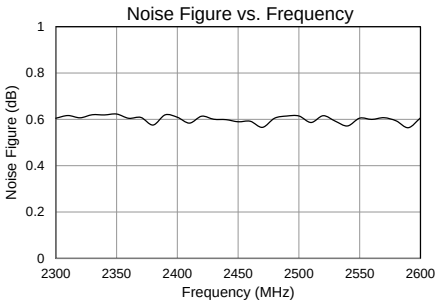



## Performance Plots: 2300 – 2600 MHz Reference Design



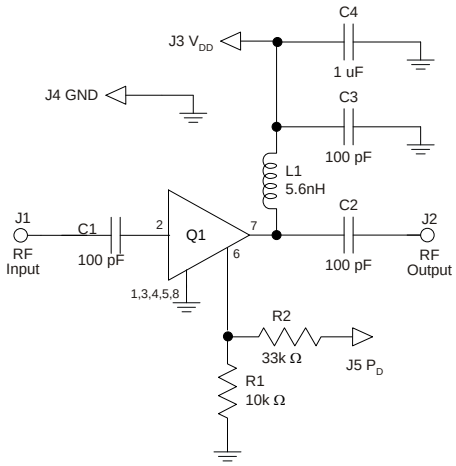



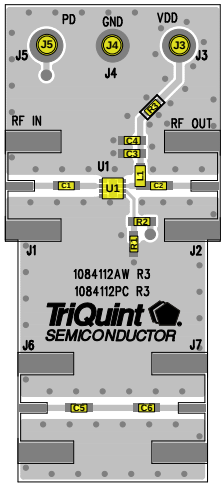




## Bill of Material: 2300 – 2600 MHz Reference Design

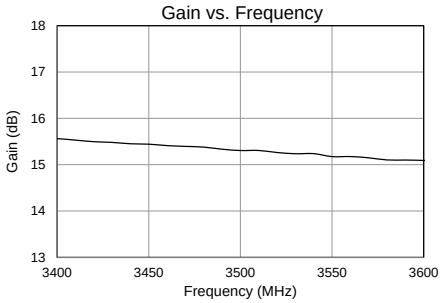
| Reference Des. | Value | Description | Manuf. | Part Number |
|---|---|---|---|---|
| n/a | n/a | Printed Circuit Board | Qorvo | |
| U1 | n/a | Ultra Low Noise Amplifier | Qorvo | TQP3M9037 |
| R1 | 10 kΩ | Resistor, Chip, 0402, 5%, 1/16W | various | |
| R2 | 33 kΩ | Resistor, Chip, 0402, 5%, 1/16W | various | |
| R3 | 0 Ω | Resistor, Chip, 0402, 5%, 1/16W | various | |
| L1 | 5.6 nH | Inductor, 0603, 5%, Ceramic | Coilcraft | 0603CS-5N6XJL |
| L2 | 2.7 nH | Inductor, 0402, 5%, Ceramic | TOKO | LL1005-FHL2N7S |
| C1 | 2.4 pF | Cap, Chip, 0402, ±0.25 pF, 50V, NPO/COG | AVX | 04023J2R4BBSTR |
| C4 | 1.0 uF | Cap, Chip, 0402, 10%, 10V, X5R | various | |
| C2, C3, C5, C6 | 100 pF | Cap, Chip, 0402, 5%, 50V, NPO/COG | various | |

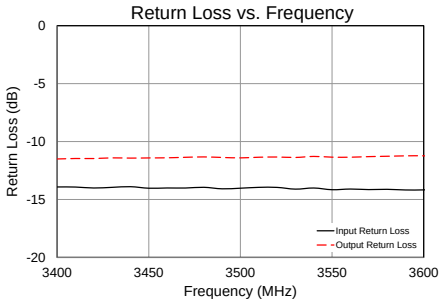
## Evaluation Board: 3400 -3600 MHz Reference Design

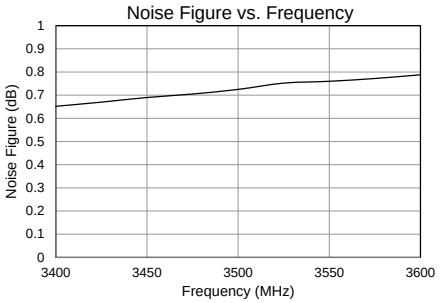



## Performance Plots: 3400 -3600 MHz Reference Design



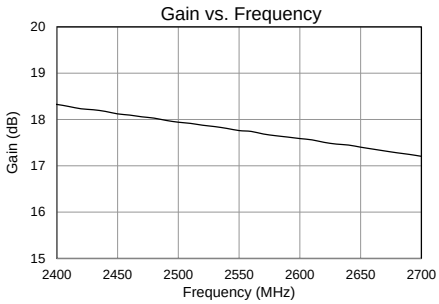

## Bill of Material: 3400 -3600 MHz Reference Design

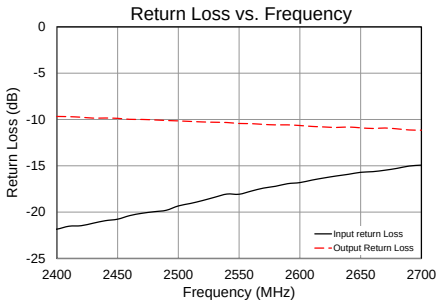
| Reference Des. | Value | Description | Manuf. | Part Number |
|---|---|---|---|---|
| n/a | n/a | Printed Circuit Board | Qorvo | |
| U1 | n/a | Ultra Low Noise Amplifier | Qorvo | TQP3M9037 |
| R1 | 10K Ω | Resistor, Chip, 0402, 5%, 1/16W | various | |
| R2 | 33K Ω | Resistor, Chip, 0402, 5%, 1/16W | various | |
| R3 | 0 Ω | Resistor, Chip, 0402, 5%, 1/16W | various | |
| L1 | 5.6 nH | Inductor, 0603, 5%, Ceramic | Coilcraft | 0603CS-5N6XJL |
| C4 | 1.0 uF | Cap, Chip, 0402, 10%, 10V, X5R | various | |
| C1, C2, C3, C5, C6 | 100 pF | Cap, Chip, 0402, 5%, 50V, NPO/COG | various | |

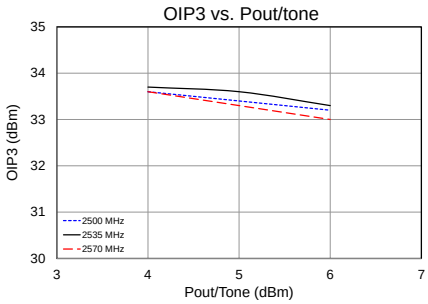
## +3.3 V $V_{DD}$ Performance Improvement with External Input Bias

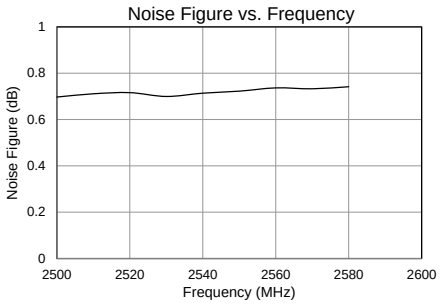
This page shows performance data of TQP3M9037 tuned for better input return loss for B7 UL. The device is biased at 3.3V Vdd. At this bias voltage the device draws about 45mA Idd but the OIP3 drops by almost 5dB. To improve the linearity performance an external bias is introduced to the input port by a resistive divider tied to the 3.3V. This results in increased current draw of about 60mA and the OIP3 is up by 5dB.

## Performance Plots

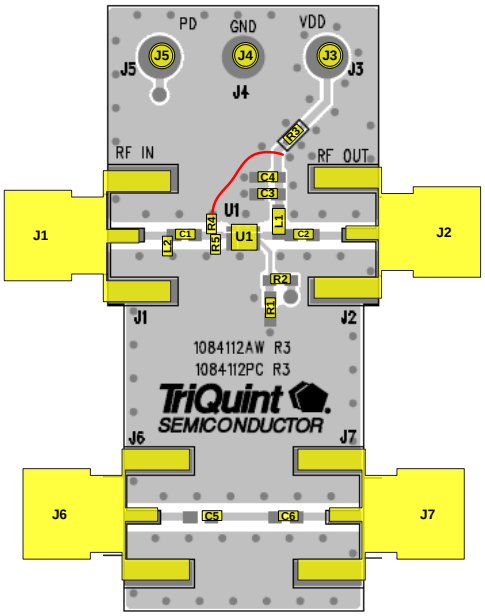


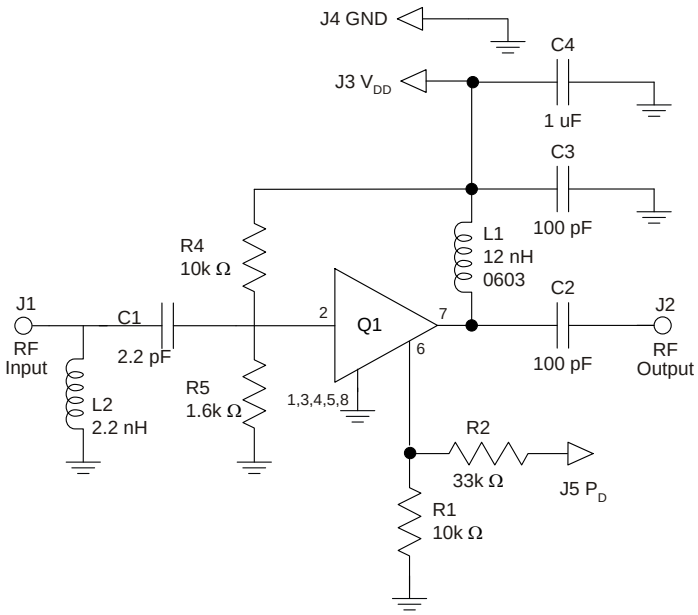




## Evaluation Board

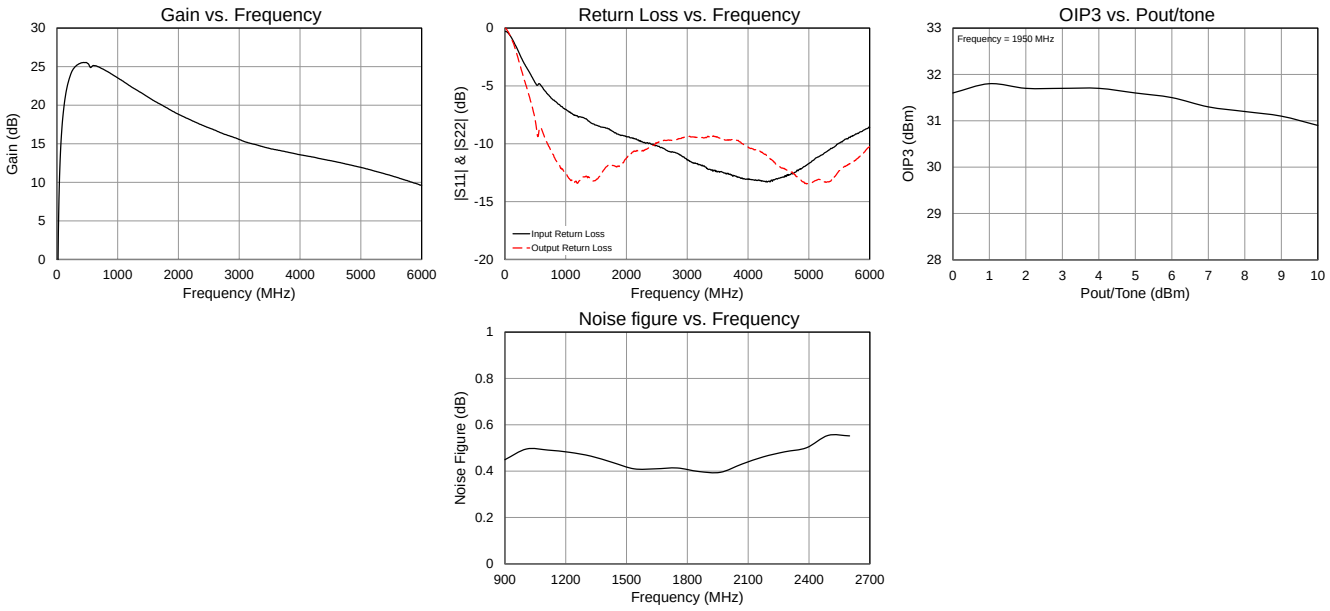


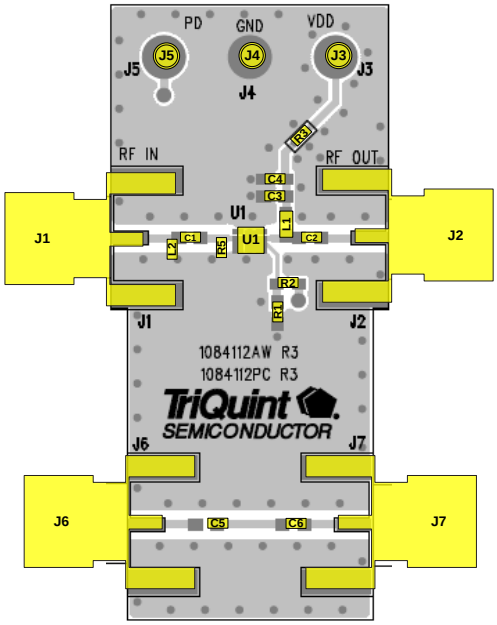
## +5 V $V_{DD}$ Reduced Current Reference Design

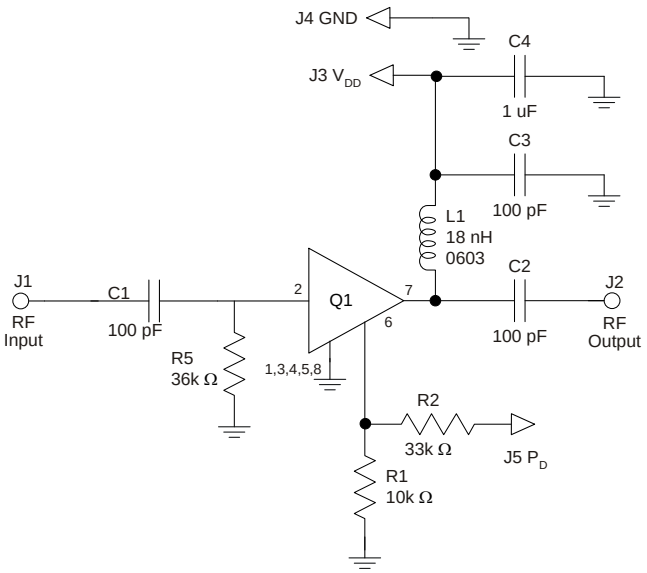
This page shows performance data of TQP3M9037 with reduced current consumption of around 40mA. The Vdd is kept at +5V and to reduce the current a shunt resistor of 36K ohms is added at the input pin. This drops the gate voltage of the input transistor and hence bringing the bias point of the LNA down.

## Performance Plots

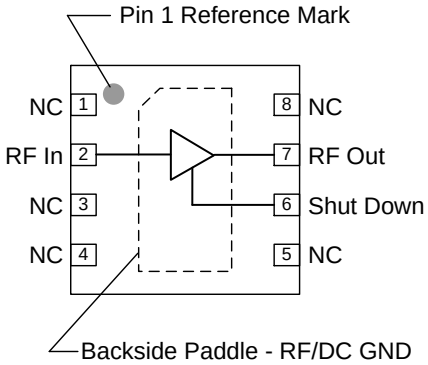



## Evaluation Board



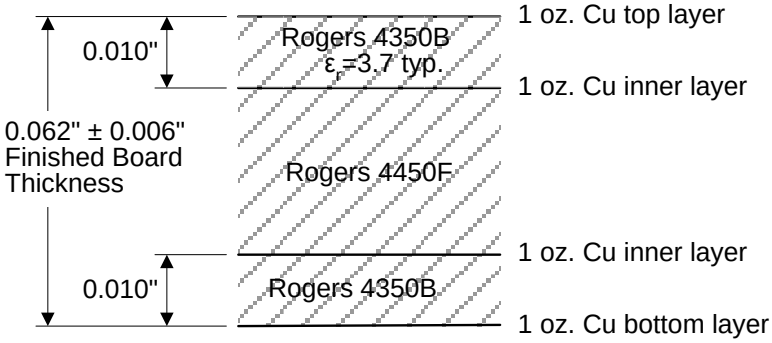
## Pin Configuration and Description



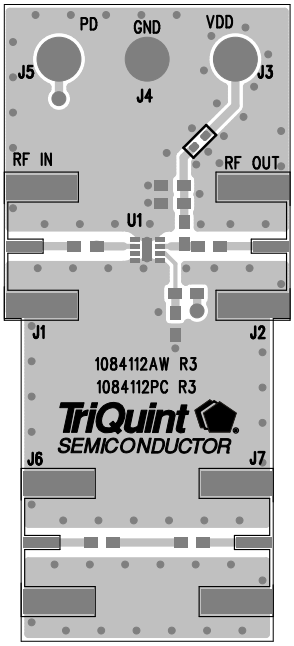

| Pin No. | Label | Description |
|---|---|---|
| 2 | RF In | RF Input pin. A DC Block is required. |
| 6 | Shut Down | A high voltage turns off the device. If the pin is not connected or is less than 1V, then the device will operate under its normal operating condition. |
| 7 | RF Out / DCBias | RF Output pin. DC bias will also need to be injected through a RF bias choke/inductor for operation. |
| 1, 3, 4, 5, 8 | NC | No electrical connection. Provide grounded land pads for PCB mounting integrity. |
| Backside Paddle | RF/DC GND | RF/DC ground. Use recommended via pattern to minimize inductance and thermal resistance; see PCB Mounting Pattern for suggested footprint. |

## Evaluation Board PCB Information

Qorvo PCB 1084112 Material and Stack-up



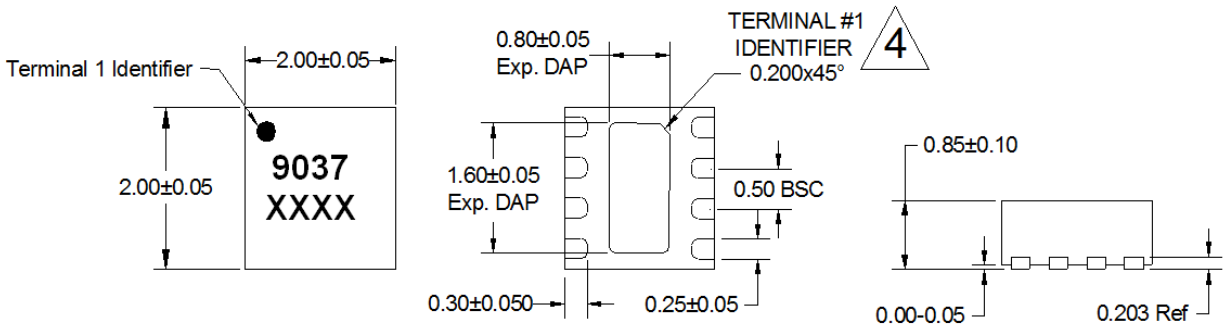

50 ohm line dimensions: width = 0.020”, spacing = 0.032”



## Mechanical Information

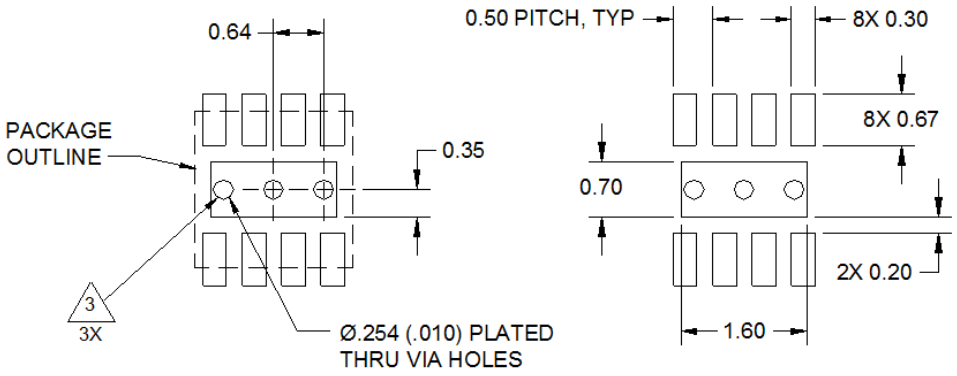
### Package Marking and Dimensions

Marking: Part number – 9037
Lot code – XXXX




Notes:
1. All dimensions are in millimeters. Angles are in degrees.
2. Except where noted, this part outline conforms to JEDEC standard MO-220, Issue E (Variation VGGC) for thermally enhanced plastic very thin fine pitch quad flat no lead package (QFN).
3. Dimension and tolerance formats conform to ASME Y14.4M-1994.
4. The terminal #1 identifier and terminal numbering conform to JESD 95-1 SPP-012.

## PCB Mounting Pattern




Notes:
1. All dimensions are in millimeters. Angles are in degrees.
2. Use 1 oz. copper minimum for top and bottom layer metal.
3. Vias are required under the backside paddle of this device for proper RF/DC grounding and thermal dissipation. We recommend a 0.35mm (#80/.0135") diameter bit for drilling via holes and a final plated thru diameter of 0.25 mm (0.10”).
4. Ensure good package backside paddle solder attach for reliable operation and best electrical performance.

## Handling Precautions

| Parameter | Rating | Standard |
|---|---|---|
| ESD – Human Body Model (HBM) | Class 1B | ESDA / JEDEC JS-001-2014 |
| ESD – Charged Device Model (CDM) | Class C2 | ESDA / JEDEC JS-002-2014 |
| MSL – Moisture Sensitivity Level | 1 | IPC/JEDEC J-STD-020 |



Caution!
ESD-Sensitive Device

## Solderability

Compatible with lead-free (260°C max. reflow temp.) soldering process.
Solder profiles available upon request.

Contact plating: NiPdAu

## RoHS Compliance

This part is compliant with 2011/65/EU RoHS directive (Restrictions on the Use of Certain Hazardous Substances in Electrical and Electronic Equipment) as amended by Directive 2015/863/EU.

This product also has the following attributes:

- Lead Free
- Halogen Free (Chlorine, Bromine)
- Antimony Free
- TBBP-A ($C_{15}H_{12}Br_4O_2$) Free
- PFOS Free
- SVHC Free




## Contact Information

For the latest specifications, additional product information, worldwide sales and distribution locations:

**Web:** www.qorvo.com

**Tel:** 1-844-890-8163

**Email:** customer.support@qorvo.com

For technical questions and application information: **Email:** appsupport@qorvo.com

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